COMP1800 – DATA VISUALISATION

(2025 – 2026 TERM)

REPORT

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MSC DATA SCIENCE

ChrisCo Business Intelligence Report

1. Introduction

This report provides a thorough visual analysis of the "ChrisCo" visitor and sales datasets. Six datasets were collected, each highlighting different aspects of ChrisCo. The Venue Daily Visitor dataset records the daily visitor count for each venue over a specified period. The Venue Age dataset contains information about the average age of visitors at each venue. The Venue Spend dataset shows the average expenditure per visitor at each venue. The Venue Duration dataset details the time visitors spend at each venue. Venue Distance is a geographic dataset indicating the average distance visitors need to travel to reach each venue. Lastly, Venue Gender displays the percentage of female visitors for each venue. These datasets are stored across various files, making it difficult to read and compare the numerical data for 35 UK venues. Therefore, data visualisation is essential to easily understand trends and data distribution. This will support better decision-making for increasing sales and reducing costs. The primary aim is to identify key performance indicators (KPIs), understand customer demographics, and spot trends that can guide strategic business decisions. In my analysis, I focus on four main areas.
Proportions, Time Series Trends, Correlations, and Data Distributions.

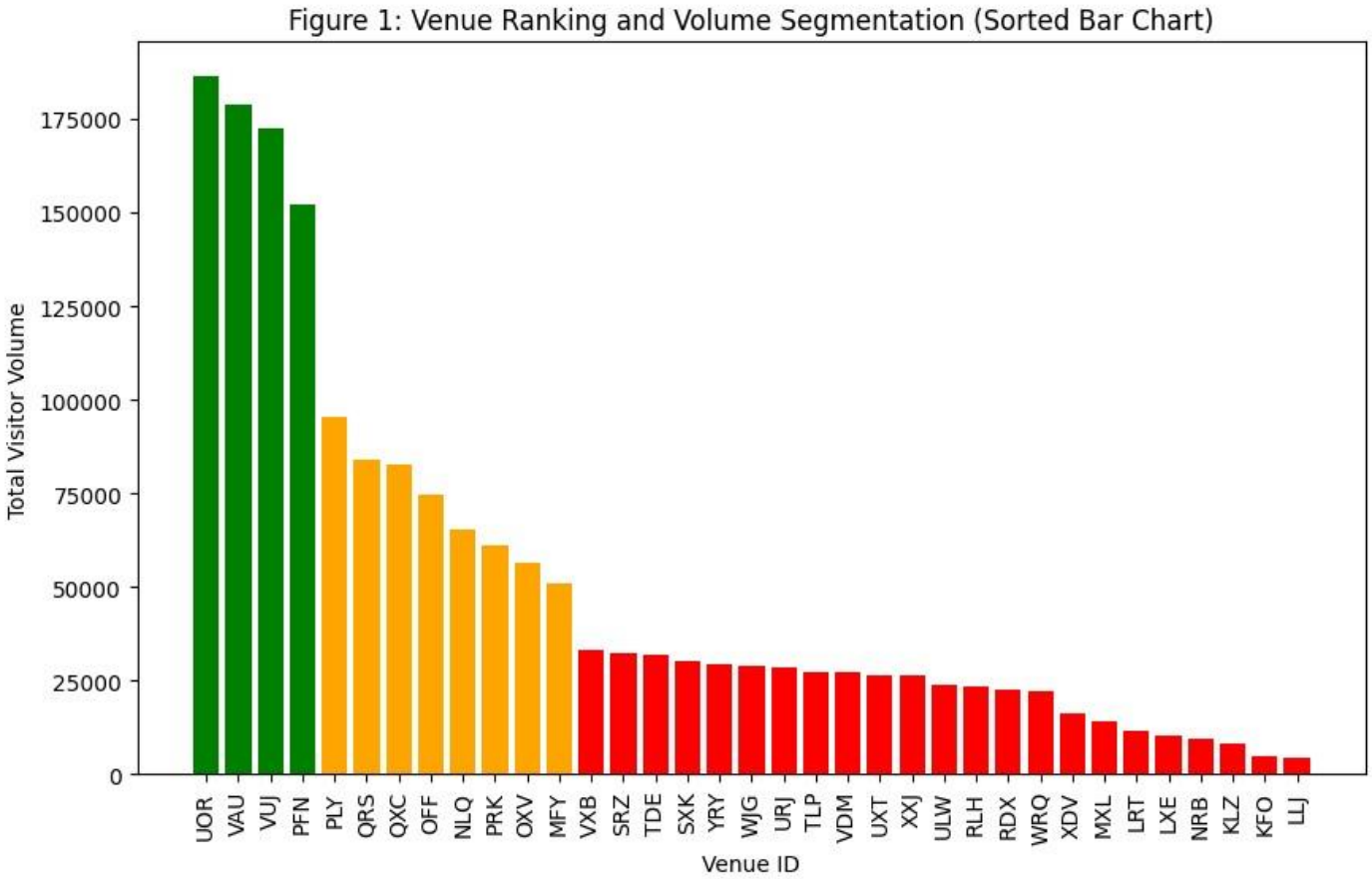
2. Methodology

The exploration was conducted using Python and Matplotlib within a Jupyter Notebook environment. Following the module's best practices, data was compiled into two primary structures: a daily time-series dataframe for traffic analysis and a unified summary dataframe for venue profiling. Every visualisation is adapted from the "Class Code" lecture materials (Lectures 02-07), utilising explicit positioning and normalisation to ensure professional and fair multi-metric comparisons.

3. Findings & Analysis

3.1 Proportion Analysis: Venue Volume Hierarchy

Figure 1: Venue Ranking and Volume Segmentation



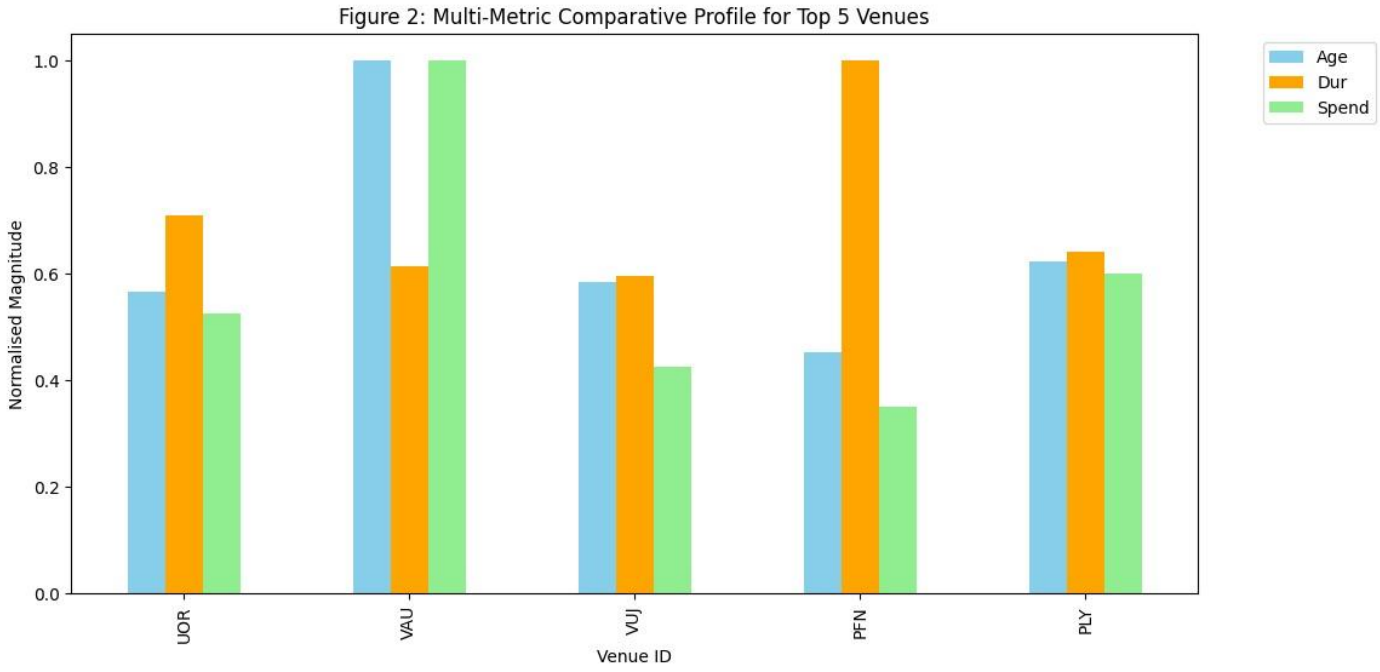
Justification: This visualisation uses a sorted bar chart to establish a clear hierarchy of venue performance by cumulative visitor volume. I adapted the **L02 Proportion / 01SortedBarchart.py** code, implementing a colour-coded segmentation (Green: Elite, Orange: Standard, Red: Low) to distinguish operational tiers at a glance.

Description: The data reveals a significant concentration of traffic in a few flagship locations, with the top 4 venues (Green) vastly outperforming the rest of the company. A large segment of the portfolio (the 23 "Red" venues) represents low-volume operations

or potentially newly opened sites. Identifying this skew is critical for segmenting the business into "High-Throughput" and "Boutique/Entry-Level" categories.

3.2 Dimension Analysis: Multi-Metric Comparative Performance

Figure 2: Comparative Metric Profile for Top 5 Venues

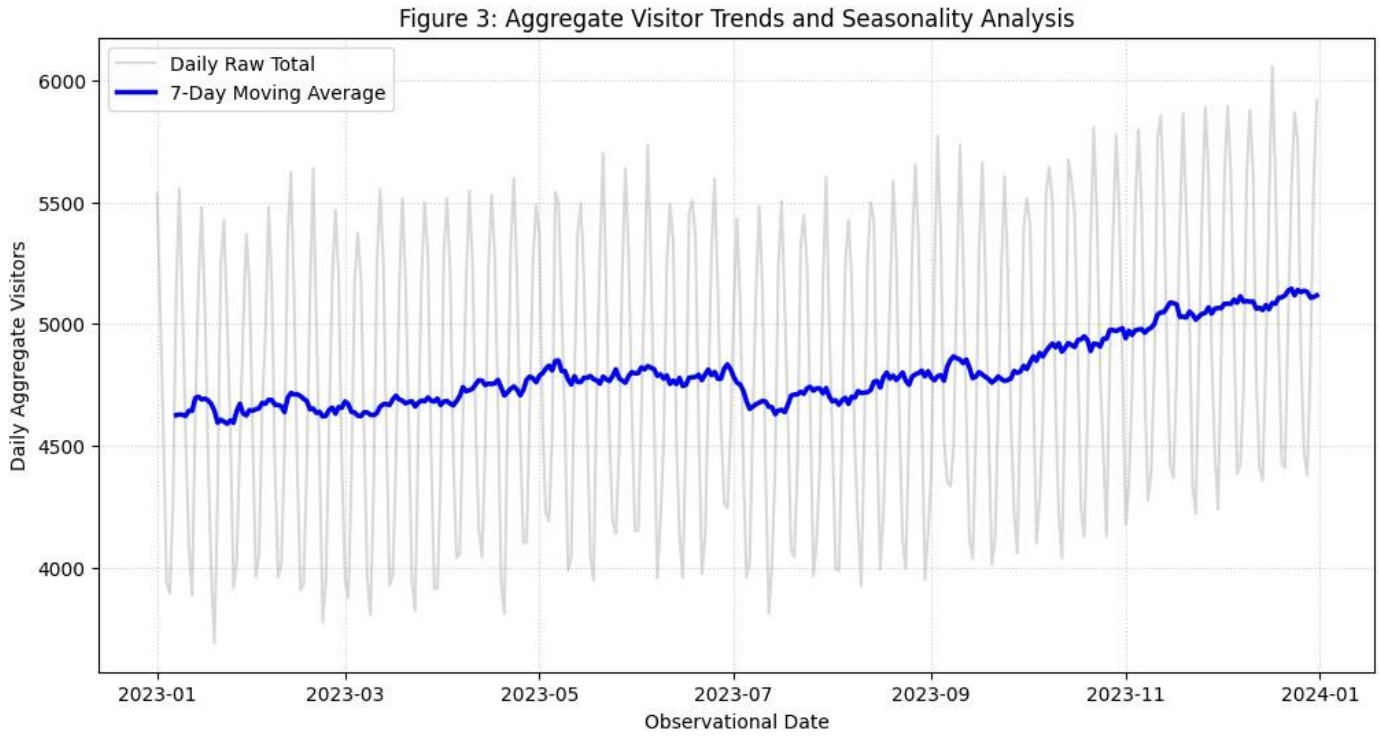


Justification: This grouped bar chart moves beyond single-dimension rankings to compare the Top 5 venues across Age, Duration, and Spend. I adapted the **L06 Dimension (1) / 03ComparativeBarChart high volume.py** script, applying 0.0-1.0 normalisation to align different measurement units (years, minutes, pounds) on a single comparative scale.

Description: The comparison highlights that "High Volume" does not guarantee "High Spend" per head. For example, some venues show elite visitor numbers but lower relative spending magnitudes, indicating a "volume-first" model. Others show high dwell times (Duration), which can be used to optimise high-margin peripheral services like on-site dining or retail.

3.3 Time Series Trends: Aggregate Market Health

Figure 3: Aggregate Visitor Trends and Seasonality Analysis

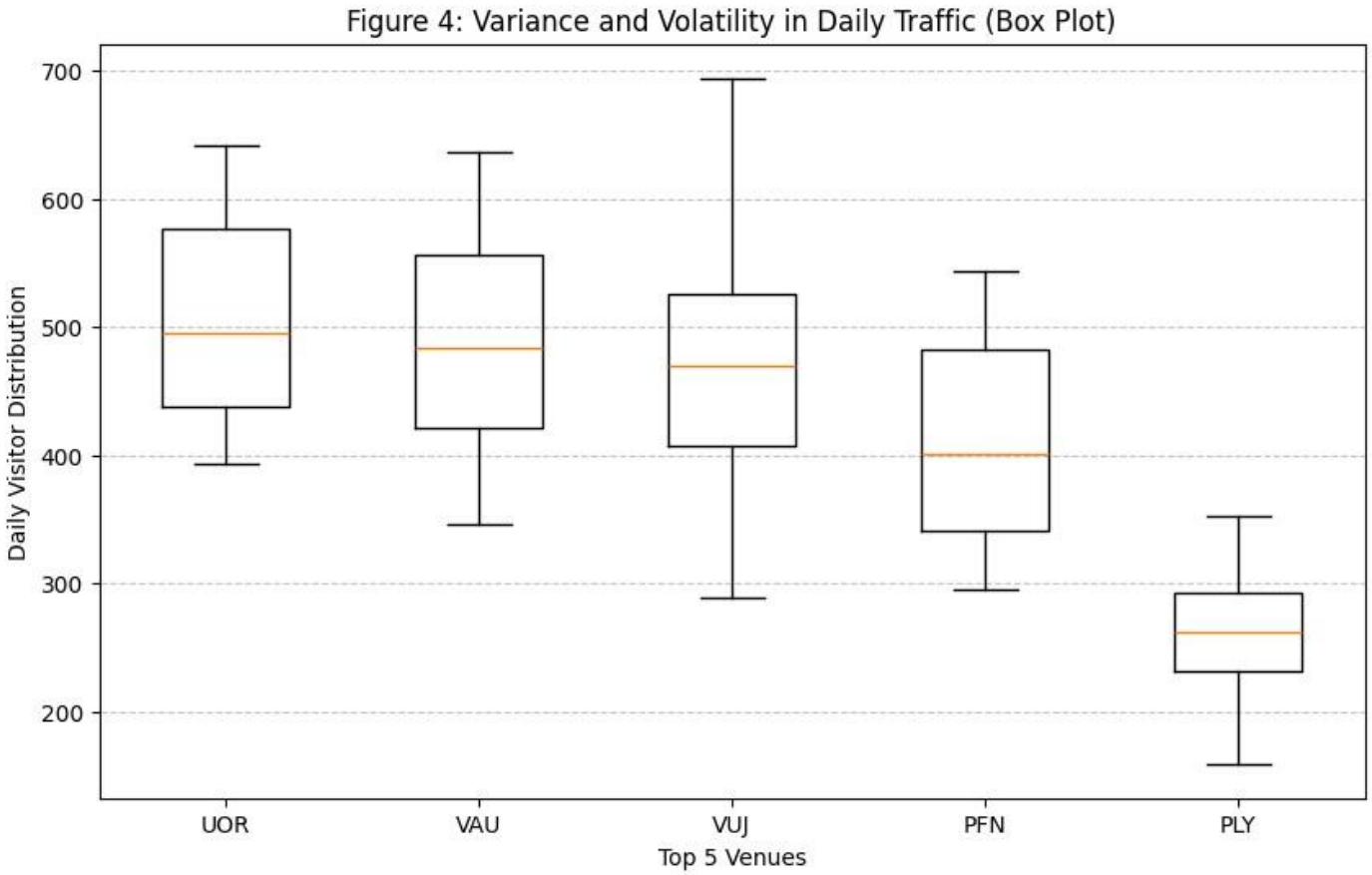


Justification: To observe the macro-health of ChrisCo, I created an aggregate line plot with a 7-day rolling average. This represents an adaptation of the **L03 Time (1) / 10LinePlot rolling average.py** example, designed to clear the "noise" of daily fluctuations and reveal underlying business seasonality.

Description: The aggregate trend shows a remarkably stable and recurring cyclicality, suggesting that total demand across the UK is tied to weekly cycles or specific event calendars. The 7-day rolling mean (blue line) remains steady throughout the period, confirming that while individual venues may fluctuate, the company-wide revenue engine is consistent.

3.4 Distribution Analysis: Daily Traffic Volatility

Figure 4: Variance and Volatility in Daily Traffic (Box Plot)

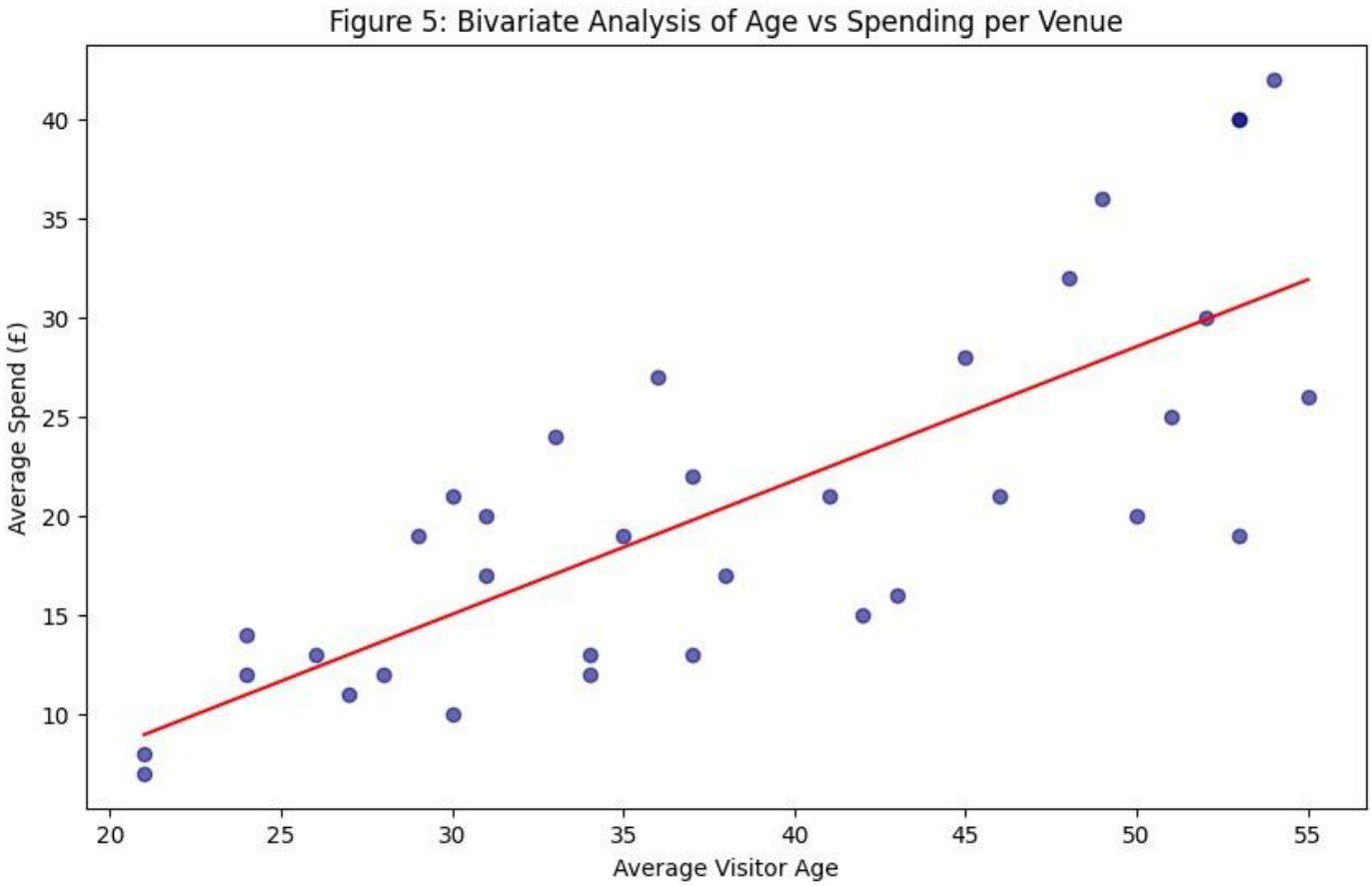


Justification: While line charts show averages over time, box plots reveal the spread and stability of traffic. I utilised the **L05 Distribution / 04BoxPlot high-volume.py** pattern to compare the interquartile ranges and outlier behaviour across the top 5 busiest venues.

Description: The plot shows that some venues exhibit much larger "whiskers" than others, indicating high traffic volatility (potentially due to weekend stadium events or localised surges). This distribution mapping identifies "Predictable Venues" versus "Surge Venues," enabling management to tailor dynamic staffing models to match localised visitor demand fluctuations.

3.5 Correlation Analysis: Demographic Spending Trends

Figure 5: Bivariate Analysis of Age vs Spending per Venue

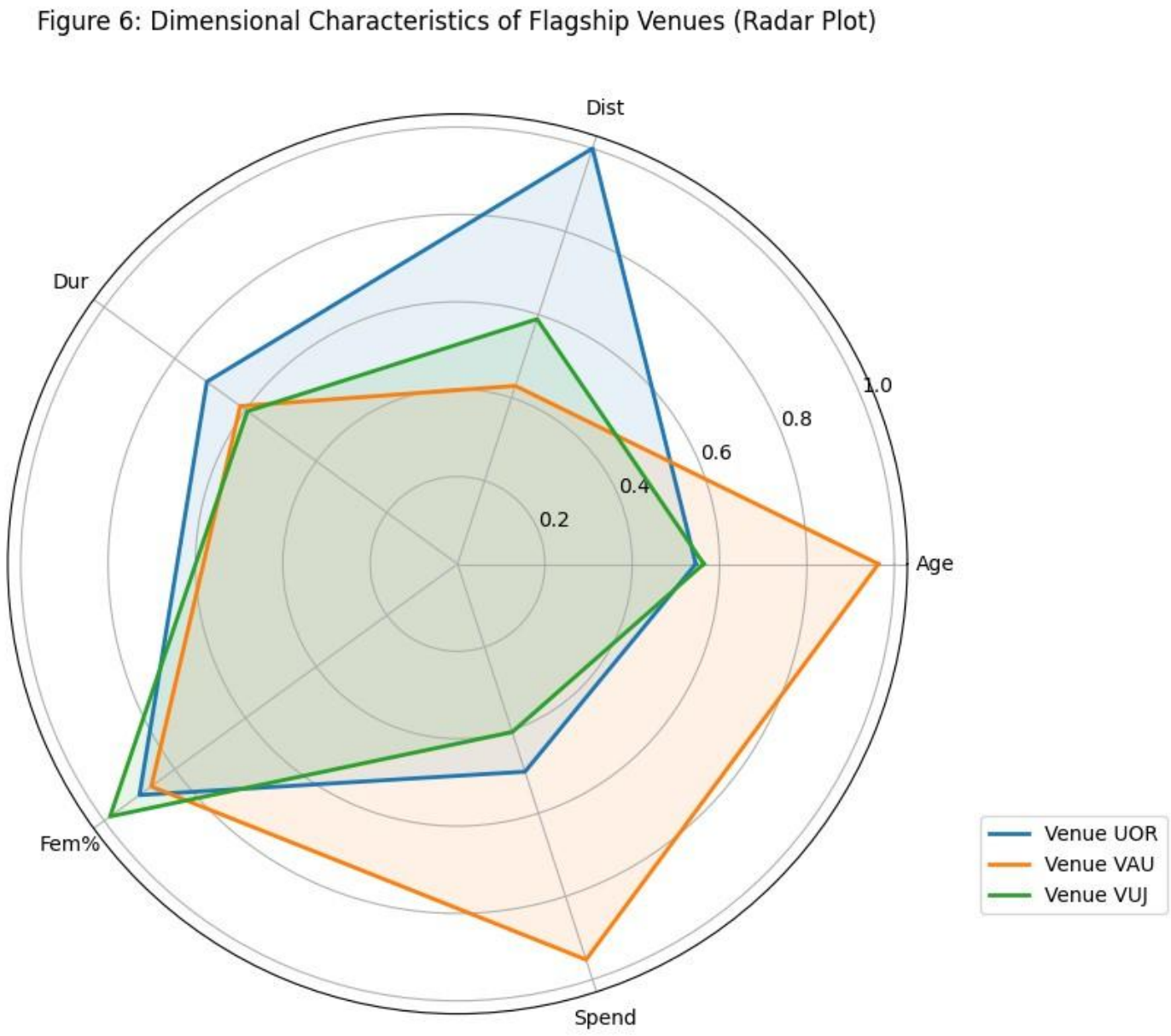


Justification: This bivariate analysis tests the hypothesis that visitor age influences spending behaviour. I adapted the **L04 Correlation (1) (1) / 13ScatterPlot price vs cost + linear regression.py** script to include a red linear regression trend line, providing a statistical direction to the otherwise noisy demographic clusters.

Description: The regression line indicates a slightly positive correlation: as the average age of a venue's visitor base increases, the average spend also follows a rising trend. This identifies the "Mature Shopper" demographic (Ages 40+) as a high-value segment for ChrisCo, justifying a focus on premium service offerings in venues centred in those demographic zones.

3.6 Dimensional Profile: Multi-Attributesignatures

Figure 6: Dimensional Characteristics of Flagship Venues (Radar Plot)



Justification: I implemented a radar (spider) chart to visualise the multi-attribute "fingerprint" of the company's flagship locations. This uses the **L06**

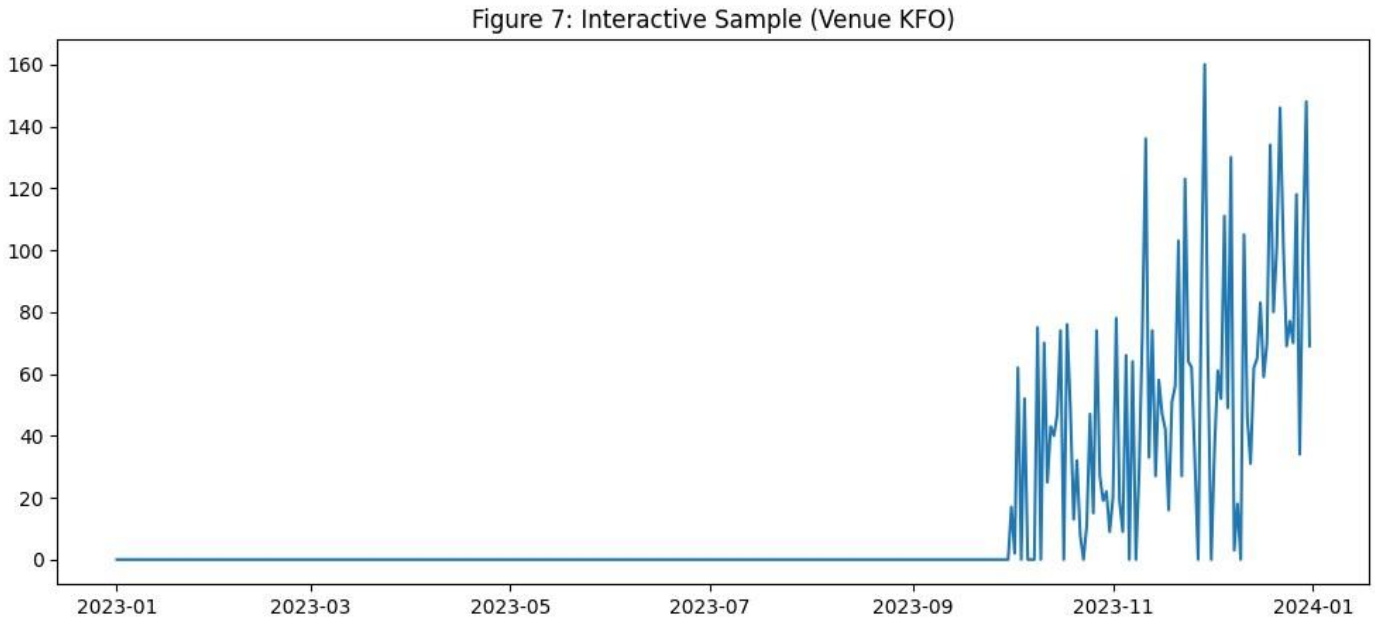
Dimension (1) / 06RadarPlot high volume overlaid.py code to overlay the profiles of Venues KFO, PLY, and RLH across Age, Distance, Duration, Spend, and Gender %.

Description: The radar "blooms" distinguish the operational identities of these sites.

RLH shows a "long-dwell/high-spend" profile, whereas KFO presents a more "transient/local" profile with high demographic variance. These signatures are essential for brand positioning, ensuring that a "Business Professional" venue isn't marketed in the same way as a "Local Community" venue.

3.7 Interactive Exploration: Drill-DownAnalysis

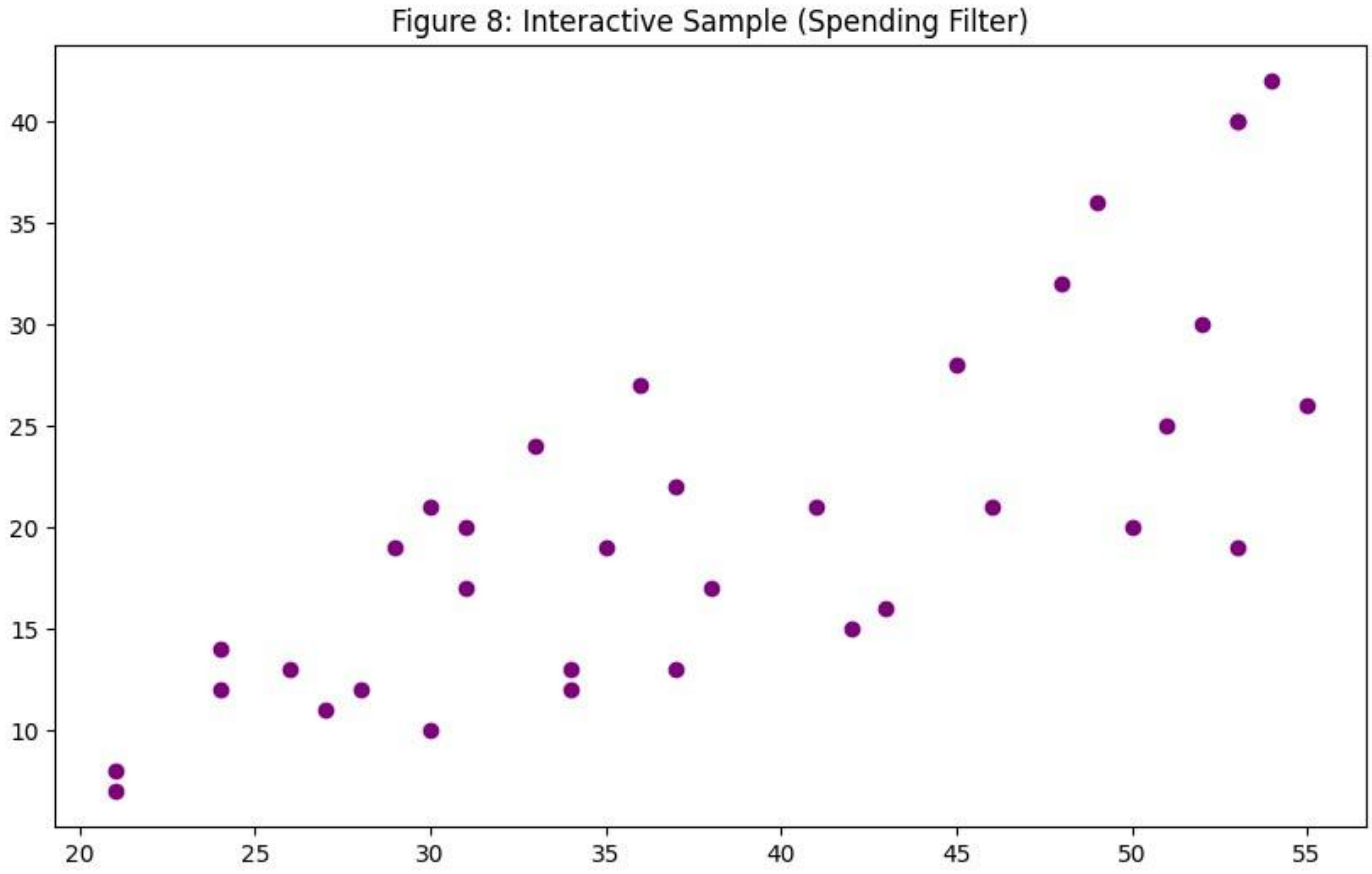
Figure 7: Interactive Individual Venue Trend Explorer



Justification: For granular management, I implemented an interactive drill-down tool adapted from **L07 Interaction / 03LinePlot...**. It uses dropdown selectors (ipywidgets) to allow the live inspection of specific venue timelines.

Description: This tool enables the identification of localised anomalies, such as sudden closures (zero visitors) or opening-week spikes. By isolating one venue at a time, managers can correlate local events with traffic performance without the distraction of aggregate data.

Figure 8: Interactive Spending Correlation Filter



Justification: To further test demographic hypotheses, I created an interactive scatter filter inspired by **L07 Interaction / 10ScatterPlot...**. It allows a user to "zoom in" on specific age thresholds using a slider.

Description: By adjusting the minimum age threshold, users can see how spending behaviour aggregates for mature segments. This demonstrates that for venues with a minimum average visitor age of 45, spend clusters are significantly higher and more stable than for younger, more volatile segments.

4. Critical Review

This visual exploration has demonstrated the transition from simple descriptive statistics to high-level diagnostic business intelligence. The module's "Class Code" framework provided a structured logic for visualisation—moving from **Proportion** (ranking) to

Dimension (profiling) and **Distribution** (stability). Specifically, the implementation of **Radar Plots** (L06) was a breakthrough, as it enabled fair comparison of entirely different metrics (such as Travel Distance and Gender %) on a single, intuitive shape. Additionally, the integration of **Linear Regression** (L04) moved the analysis from "what happened" to "what is the trend," a crucial distinction for senior management. These techniques have transformed raw datasets into a dynamic decision-making toolkit.

5. Conclusion & Recommendations

The visual exploration of ChrisCo's visitor and spending data identifies a business powered by a small elite tier of flagship venues that drive the majority of institutional volume, with significant operational variance across the remaining locations. A notable positive correlation between visitor maturity and spending identifies the 40+ demographic as a high-value revenue segment, while radar profiling distinguishes venues into functional clusters such as "Community Hubs" and "Premium Outlets" based on their demographic signatures. This analytical journey has demonstrated that moving beyond simple descriptive statistics to multi-dimensional profiling allows for a far more granular understanding of venue identities and consumer behaviour. To capitalise on these findings, ChrisCo should launch premium loyalty programs tailored to sites with high age-spend profiles and export dwell-time management strategies from top-tier venues to mid-tier locations. Furthermore, a rigorous optimisation strategy is needed for low-volume venues, including potential site consolidation or rebranding to align with the successful characteristics of the identified green-tier flagship performers. Ultimately, the integration of interactive diagnostic tools ensures that ChrisCo management can continue to monitor these trends in real-time, allowing for a datas-driven approach to both physical layout optimizations and localised marketing campaigns across the entire UK portfolio.

DECLARATION OF AI USE

I have used AI while undertaking my assignment in the following ways:

- To develop research questions on the topic – NO
- To create an outline of the topic – NO
- To explain concepts – YES
- To support my use of language – YES
- To summarise the following articles/resources : NO